

Cirrus NCR

ARCHER2 Transition Information

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Overview



- National Compute Resources (NCRs)
- ARCHER2 → Cirrus NCR migration
- Differences from and similarities to ARCHER2
 - Hardware
 - Software
- Help and support

<https://docs.cirrus.ac.uk/archer2-migration-2026/>



Mr. Alejandro de la Calle

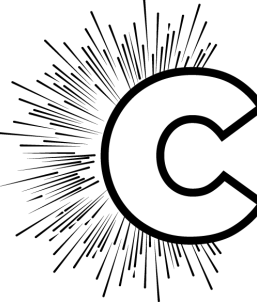
National Compute Resources (NCR)



- NCRs will complement the next national supercomputing service
 - Provide a portfolio of diverse technologies available to the UK's research and innovation communities.
 - NCRs will collaborate closely with each other
- UKRI anticipate a rolling programme of phased investment in NCRs, informed by unmet technological requirements in the ecosystem.
- The first phase of NCRs is launching through the second half of 2026 to early 2027
 - Cirrus NCR is one of the systems that is already available.



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 **cirrus**

- HPE Cray EX4000 supercomputer
- 184,320 cores, 640 compute nodes
 - AMD EPYC 9845 (144c) – two per node
 - 768/1536 GB DDR5 per node
 - HPE Cray Slingshot 11 interconnect
- > 50 PB CephFS and VAST storage
- Software
 - RHEL 9
 - Slurm
 - HPE Cray Programming Environment
 - EPCC software installs (via Spack)

NCR Phase 1: CPU systems



NCR	Host	Vendor	Technical details
Cirrus NCR https://www.cirrus.ac.uk	EPCC	HPE	184,320 cores (640 nodes). Each node: • 2x AMD EPYC 9825 (144c) • 768/1536 GB DDR5 • 200 Gbps dual rail Slingshot 11 interconnect
Charger NCR	UCL	HPE	37,888 cores (148 nodes). Each node: • 2x Intel 6980P (128c) • 1536 GB DDR5 • 400 Gbps Slingshot 11 interconnect
Isambard 3 NCR https://docs.isambard.ac.uk/	U of Bristol	HPE	54,720 cores (380 nodes). Each node: • 2x NVIDIA Grace-Grace (72c) • 120 GB memory per Grace CPU • 200 Gbps single rail Slingshot 11 interconnect

NCR Phase 1: GPU systems

NCR	Host	Vendor	Technical details
Baskerville NCR https://www.baskerville.ac.uk/	University of Birmingham	Lenovo	22 nodes. Each node: • 8x NVIDIA B200 GPU – each with 180 GB HBM • 12 TB local SSD • 4x 400 Gbps NDR Infiniband
Mary Coombs NCR https://www.hartree.stfc.ac.uk/technologies/mary-coombs/	STFC Daresbury Lab	Lenovo	157 nodes. Each node: • 4x NVIDIA H100 GPU – each with 80 GB HBM • 4x 200 Gbps NDR Infiniband
Zenith NCR https://docs.hpc.cam.ac.uk/user-guide/zenith-quickstart.html	University of Cambridge	Dell	28 nodes. Each node: • 8x AMD Instinct MI355X GPU – each with 288 GB HBM • 30.4 TB local SSD • 8x 400 Gbps NDR Infiniband + 1x NDR dedicated to storage

Christopher Ellis

ARCHER2 → Cirrus NCR Migration

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Project/account management and setup



- Cirrus projects and accounts managed through SAFE in the same way as you do for ARCHER2
- Adding Cirrus access for migrating projects from **16 Sep 2026** onwards
 - Many projects already setup – working through the rest of the list
 - You will usually keep the same project code as you had on ARCHER2
- Once Cirrus has been added to a project, users in that project can request accounts through SAFE in the usual ways, either:
 - Add Cirrus to existing account (if your account already has access to NSCDS/EPCCFS storage on ARCHER2)
 - Request a Cirrus account in the project with the **same username** as on ARCHER2
- Project leads manage Cirrus resources (coreh allocations and storage quotas) through SAFE

<https://docs.cirrus.ac.uk/archer2-migration-2026/account-migration/#how-do-i-set-up-an-cirrus-ncr-account>

Project/account management (consortia) | epcc |

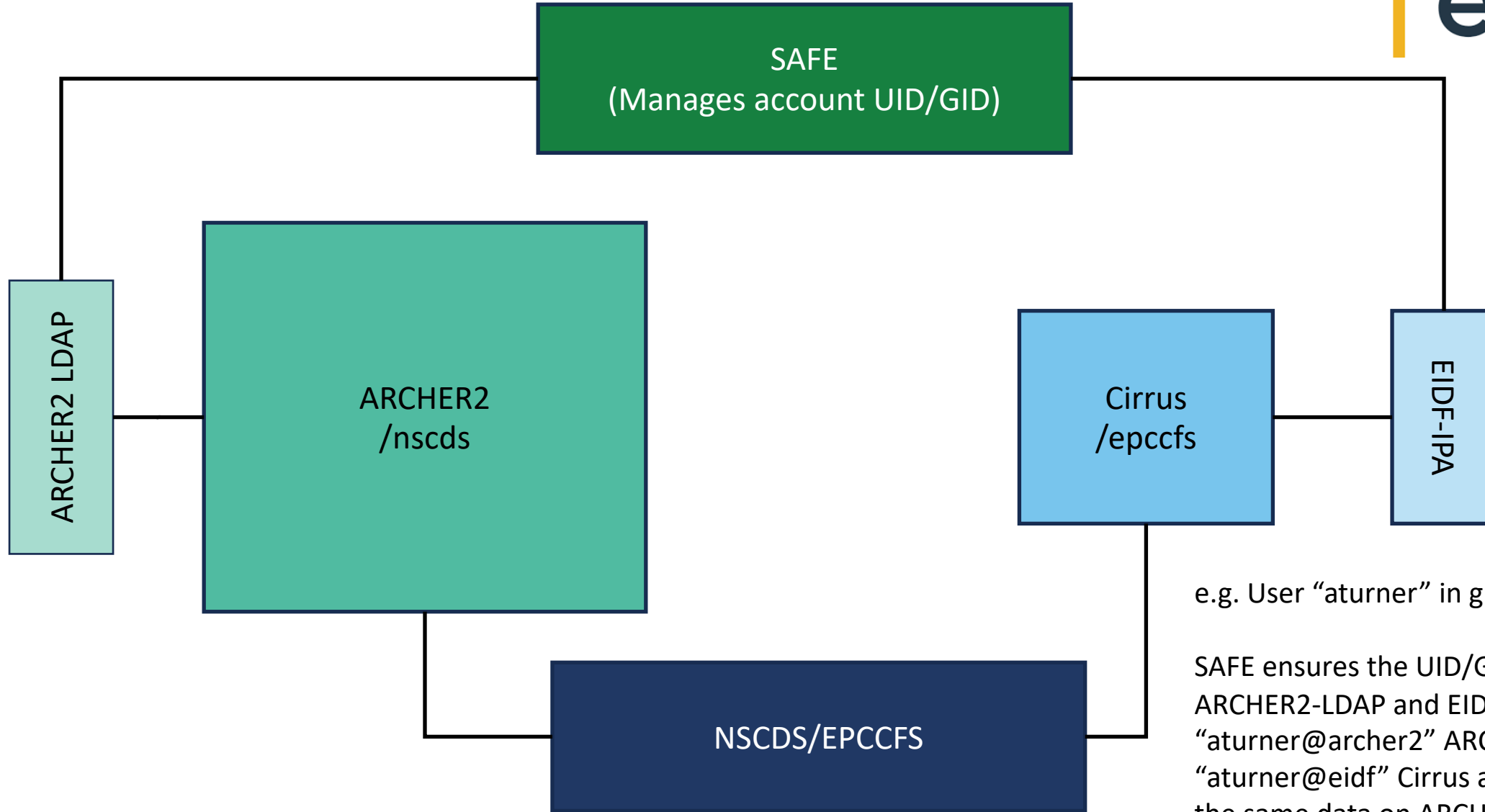
- EPSRC HEC will continue to be managed under a single code on Cirrus as they are on ARCHER2 (e.g. MCC as e05)
- NERC consortia will continue to be managed under a single code on Cirrus as they are on ARCHER2 (e.g. Atmospheric and Polar Science as n02)
- Once Cirrus has been added to a consortium, users in that consortium can request accounts through SAFE in the usual ways
- Consortium leads manage Cirrus resources (coreh allocations and storage quotas) through SAFE

Accessing Cirrus



- CLI SSH access:
 - Same as for ARCHER2 (SSH key + MFA) – users need to upload an SSH key and setup TOTP, it is not the same as for their ARCHER2 account
`ssh username@login.cirrus.ac.uk`
- Open OnDemand web access:
 - Access Cirrus via a web browser: currently supports CLI in web browser, Slurm job submission app, Jupyter Python notebooks
<https://ood.cirrus.ac.uk>
- PUMA2 (for Met Office UM workflows):
 - In process of getting this setup – will be available before end of ARCHER2 service

- Copy data locally on ARCHER2 to NSCDS/EPCCFS
 - This is the same storage system as is available on Cirrus
 - Can use local `rcclone` parallel copy to increase performance for large datasets
 - Can put in serial script for long-running copy processes
 - **Simplest and best performing option**
 - (If you do not already have access to NSCDS/EPCCFS on ARCHER2, you will gain access as soon as you create your account on Cirrus)
- Copy data using scp/sftp
 - Can use rclone parallel copy to manage transfer, increase performance
 - Can use rsync to manage transfer
 - Need to setup SSH key to access Cirrus from ARCHER2 (or vice versa) and deal with MFA
- Still working to setup Globus Online on Cirrus and aiming to have it available as soon as possible



e.g. User “aturner” in group “z19”

SAFE ensures the UID/GID is the same on ARCHER2-LDAP and EIDF-IPA so both the “aturner@archer2” ARCHER2 account and “aturner@eidf” Cirrus account can access the same data on ARCHER2 (at /nscds) and Cirrus (at /epccfs)

rclone simple example

- Example copying directory “results-data” from ARCHER2 /work to NSCDS on an ARCHER2 login node:

```
module load rclone
```

```
rclone copy /work/z19/z19/aturner/results-data /nscds/z19/z19/aturner/results-data
```

- By default, rclone will run 4 transfers in parallel, you can change this using the “--transfers <n>” option:

```
rclone --transfers 8 copy /work/z19/z19/aturner/results-data /nscds/z19/z19/aturner/results-data
```

Dr Sam Azadi

Differences from and similarities to ARCHER2

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Hardware differences



	ARCHER2	Cirrus
Processors	2x AMD EPYC 7742	2x AMD EPYC 9825
Cores per socket (node)	64 (128)	144 (288)
Cores per NUMA region	16	36
Memory per node	256/512 GB	768/1536 GB
Memory per core	2.0/4.0 GB	2.66/5.33 GB
Memory channels per socket	8	12
Memory type	DDR4	DDR5
Max. memory BW per socket (node)	190.7 (381.4) GB/s	614 (1228) GiB/s
Max. memory BW per core	3.0 GiB/s	4.3 GiB/s
Interconnect	HPE Cray Slingshot 10	HPE Cray Slingshot 11
Internet access from compute nodes?	No	Yes

Software differences: OS and CPE



	ARCHER2	Cirrus
Operating System	SLES 15 sp4	RHEL 9.6
Compilers	Cray (CCE): 16.0.1 Gnu (GCC): 11.2.0 AMD (AOCC): 4.0.0	Cray (CCE): 19.0.0 Gnu (GCC): 14.2 AMD (AOCC): 5.0.0 Intel OneAPI: 2025.0
BLAS/LAPACK/ScaLAPACK	HPE Cray LibSci: 23.09.1.1	Cray LibSci: 25.03.0 Intel OneAPI MKL: 2025.0
FFTW	FFTW: 3.3.10.5	FFTW: 3.3.10.10 Intel OneAPI: 2025.0
MPI	HPE Cray MPICH: 8.1.27	HPE Cray MPICH: 8.1.32, 9.0.0
Parallel Debugger	gdb4hpc: 4.7.3 valgrind4hpc: 4.15.1 Arm Forge (DDT): 25.1	gdb4hpc: 4.16.4 valgrind4hpc: 2.13.5 Arm Forge (DDT): 25.1
Profiler	CrayPAT: 23.09 Arm Forge (MAP): 25.1	CrayPAT: 25.03.0 Arm Forge (MAP): 25.1
Tools	HPE Cray Python: 3.10.10 HPE Cray R: 4.2.1.2	HPE Cray Python: 3.11.7 HPE Cray R: 4.4.0

Software: EPCC installations



- Use Spack for installs
 - Recommend users use this too:
<https://docs.cirrus.ac.uk/software-tools/spack/>
 - Can be used to simplify dependency installs even if the final software is not supported in Spack
- Current research software installs:
 - CASTEP, CP2K, GROMACS, LAMMPS, OpenFOAM (com/org), ORCA, Quantum Espresso, VASP 6
 - Eigen, Metis/ParMetis, PETSc, PyTorch

Slurm setup



- Very similar to ARCHER2 but at a much smaller scale
- Exception is that jobs are non-exclusive by default
 - Jobs can share nodes with other user's jobs
 - If you use “--exclusive” option for jobs smaller than a node, you will be charged for the full node
 - We strongly recommend that multi-node jobs use full nodes and specify exclusive access with the “--exclusive” sbatch/salloc/srun option
- No dedicated serial nodes like ARCHER2 but you can submit single core jobs, and all compute nodes have internet access

<https://docs.cirrus.ac.uk/user-guide/batch/#quality-of-service-qos>

Storage systems

- Similar to ARCHER2
- /home – data not needed by jobs at runtime
 - Not available on compute nodes
 - Hosted on [CephFS](#)
 - No backup
- /epccfs – data needed by jobs (i.e. similar to ARCHER2 /work)
 - Available on login and compute nodes
 - Parallel NFS file system hosted on [VAST](#)
 - No backup
 - Snapshots of previous file versions held for up to 5 weeks
- Unlike ARCHER2 /work Lustre – parallel IO performance scales well within a single node
 - Particularly useful for software that use parallel IO servers to manage IO

Dr Alfonso Bueno Orovio

Help and support

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lepcci

Getting help

- Cirrus documentation: <https://docs.cirrus.ac.uk/>
- Migration from ARCHER2: <https://docs.cirrus.ac.uk/archer2-migration-2026/>
- Cirrus service desk: support@cirrus.ac.uk
- (Beta) Cirrus AI Skills: <https://skills.cirrus.ac.uk/>
 - Continuing to develop these

Dr. Marco Rosti

Summary

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Summary



- Cirrus is much smaller than ARCHER2 – expect more contention for jobs and potentially longer queue times
- Projects migrating from ARCHER2 to Cirrus will use the same SAFE to manage accounts and resources
- Users will request accounts on Cirrus in the same project code as on ARCHER2 and use the same username
- Login process is the same as ARCHER2: SSH key + TOTP
 - Can also access Cirrus in web browser via Open OnDemand
 - PUMA2 will be available for Met Office UM job management in the future
- Simplest way to transfer data is to copy to NSCDS/EPCCFS on ARCHER2 – it is then available on Cirrus straight away
- Hardware and software setup are very similar to ARCHER2
- Slurm is also similar in setup
 - Except that nodes can be shared between users and node-exclusive access needs to be explicitly requested if you want it

<https://docs.cirrus.ac.uk/archer2-migration-2026/>

ARCHER2 to Cirrus NCR Migration



<https://docs.cirrus.ac.uk/archer2-migration-2026/>